

BSCDS44: REINFORCEMENT LEARNING**BSc Data Science. IV Year VII Sem.**

Teaching Scheme: TH: - 3 Hours/Week	Credit TH: 03	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks
Course Prerequisites: Basic knowledge of probability and statistics as well as all concepts of machine learning		
Course Objective: Understand the fundamentals of reinforcement learning (RL) algorithms. Learn to implement various RL algorithms and understand their trade-offs. Explore applications of RL in different domains such as robotics, gaming, and finance. Gain insights into advanced topics like deep reinforcement learning. Develop the ability to critically analyze and solve RL problems.		
Learning Course Outcome: At the end of the course the students will be able to CO1: Ability to explain the basic concepts of reinforcement learning, including Markov Decision Processes (MDPs) and reward functions. CO2: Proficiency in implementing key RL algorithms such as Q-Learning, SARSA, and Policy Gradient methods. CO3: Capability to apply RL techniques to solve control and decision-making problems in simulated environments. CO4: Understanding of how to tune RL algorithms and parameters for optimal performance. CO5: Skills to design and develop RL-based solutions for real-world problems, considering ethical implications and practical constraints. CO6: Implement and deploy a Deep Q-Learning-based recommendation system to optimize driver performance and retention.		
Course Contents		
UNIT-I	Introduction to Classical Reinforcement Learning	8 Hours
Understanding Generative AI: Concepts and applications. Evolution of Language Models: From rule-based systems to large language models. Introduction to ChatGPT: Features, capabilities, and use cases. Navigating Ethical Considerations: AI-generated content and responsible usage		
UNIT-II	Classic RL Assignment Introduction: Overview of the tic-tac-toe challenge	8 Hours
Classical RL Algorithms and Tic-Tac-Toe Agent: Recap of Dynamic Programming and Monte Carlo Methods. In-depth Q Learning: Exploring the Q Learning algorithm. Training the Agent: Implementing a Q Learning agent for tic-tac-toe. Evaluation Metrics: Defining success criteria		

for the agent. Fine-Tuning Strategies: Enhancing the agent's performance. Classic RL Assignment: Developing and testing the tic-tac-toe agent.		
UNIT-III	Introduction to Deep Reinforcement Learning	8 Hours
Basic Concepts of Deep Learning: Neural networks and their applications. Merging RL and Deep Learning: Motivation and advantages. Deep Q Learning: Overview and combining Q Learning with neural networks. Policy Gradient Methods: Learning policies directly using gradient ascent. Actor-Critic Method: Combining policy-based and value-based approaches		
UNIT-IV	Deep Reinforcement Learning Algorithms Deep Q Learning in Depth	8 Hours
Working principles and training process. Policy Gradient Methods: Understanding policy gradients and their training. Actor-Critic Approach: Exploring the actor-critic architecture and training. Challenges of Deep RL: Exploring potential issues and mitigations		
UNIT-V	Reinforcement Learning Project Introduction	6 Hours
Project Scope: Enhancing cab driver recommendations using RL. Business Context: Understanding the importance of driver retention. Cab Aggregator Service: Platform overview and recommendation system.		
UNIT-VI	RL Project Implementation and Deployment	7 Hours
Vanilla Deep Q Learning (DQN): Overview and application for the project. Defining the Reward Structure: Designing rewards to maximize driver profits. Implementing DQN: Coding the algorithm for the recommendation problem. Model Evaluation: Testing and validating the RL-based recommendation system. Deployment and Impact: Deploying the solution and measuring its impact on driver retention.		
TEXTBOOKS: 1. Reinforcement Learning: An Introduction 2. Deep Reinforcement Learning Hands-On 3. Algorithms for Reinforcement Learning 4. Foundations of Deep Reinforcement Learning		